

COURSE TITLE : HEAT TRANSFER LAB
COURSE CODE : 5078
COURSE CATEGORY : B
PERIODS/ WEEK : 3
PERIODS/ SEMESTER : 39
CREDIT : 2

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Radiation constant, Steffen Boltzmann Constant	11
2	Heat Transfer in Shell and Tube Heat Exchanger	11
3	Heat Transfer coefficient in Natural Convection Heat transfer coefficient in Forced Convection	11
4	Thermal conductivity of materials	6
TOTAL		39

COURSE OUTCOMES :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	Determine Radiation constant
	2	Determine Steffen Boltzmann Constant
	3	Determine rate of heat Transfer in Shell and Tube Heat Exchanger
2	4	Determine heat Transfer coefficient in Natural Convection
	5	Determine overall heat transfer coefficient in Forced Convection
	6	Find Thermal conductivity of materials

List of Experiments

1. Determine Radiation constant
2. Determine Steffen Boltzmann Constant
3. Determine overall heat transfer coefficient in Double Pipe Heat Exchanger
4. Determine rate of heat Transfer in Shell and Tube Heat Exchanger
5. Determine heat Transfer coefficient in Natural Convection
6. Determine overall heat transfer coefficient in Forced Convection

DEMONSTRATION

Experiments

1. Know working of Single Effect Evaporator
2. Understand Multiple Effect Evaporators.
3. Determine Thickness of insulation
4. Find Thermal conductivity of materials